

Weather API Request Logger

A simple and realistic Python project that fetches live weather data using the Open-Meteo API and logs all events using industry-standard logging practices.

This project demonstrates:

- API request logging
- Console + file logging
- Error handling with logs
- Slow response warnings
- Practical, real-world logging usage
- Beginner-friendly project structure

Project Structure:

```
weather_logger/  
|  
├── logs/  
|   └── weather.log  
|  
├── src/  
|   └── weather_logger.py  
|  
└── requirements.txt
```

Features:

- INFO logs (request sent, response received, weather data)

- WARNING logs (slow response >1.5 sec)
- ERROR logs (network failure, non-200 response)
- Log file saved in logs/weather.log

API Used (Free & No Key Required):

https://api.open-meteo.com/v1/forecast?latitude={lat}&longitude={lon}¤t_weather=true

Source Code (weather_logger.py included in project instructions)

Installation:

1. Clone repository
2. pip install -r requirements.txt
3. python src/weather_logger.py
4. Check logs in logs/weather.log

Requirements:

requests

Example Logs:

2025-01-12 09:22:10 | INFO | Sending request to Weather API...

2025-01-12 09:22:11 | INFO | Weather API response received successfully

Learning Outcomes:

- Using a real API
- Logging warnings & errors

- File logging with timestamps
- Real-world monitoring simulation

Future Improvements:

- Rotating logs
- JSON logs
- Multiple cities
- Dashboard (Grafana/Streamlit)
- Store weather in CSV/DB

License:

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